

# The quantum-mechanical matrix bootstrap: a cribsheet

with the specialisations needed for purely fermionic (matrix SYK) models

## Matrix SYK bootstrap project

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*Scope.* Everything here is drawn from the project library: Han–Hartnoll–Kruthoff [1] (the founding paper), Lin–Zheng [2, 3], Lin’s TASI lectures [4], Laliberte–McPeak [5], Cho–Gabai–Sandor–Yin [6], Klebanov et al. [7], Chen [8], Chang et al. [9], Turiaci–Witten [10]. Equation numbers in brackets refer to those papers. Section 5 contains project-specific statements (marked  $\star$ ) that are derived in `docs/derivations.md` or verified numerically in this repository, not taken from the literature.

## 1 Objects

### State functional.

A linear functional  $\phi(O) = \langle O \rangle$  on operators, meant to be  $\langle E|O|E \rangle$  for an energy eigenstate (or  $\text{Tr } \rho O$  for a stationary  $\rho$ ,  $[\rho, H] = 0$ ) [TASI (100); LM (1.4)–(1.5)]. Normalisation  $\phi(\mathbf{1}) = 1$  (or  $N$ ). The bootstrap never constructs the state; it constrains  $\phi$ .

### Letters and words.

Letters are the elementary operators ( $X_I, P_I, \psi_\alpha$  for BFSS;  $\Psi_{ij}, \bar{\Psi}_{ij}$  for us). Words are ordered products. At large  $N$  the variables are *single-trace* words  $\text{tr}(O_1 \cdots O_n)$  with global-symmetry indices contracted into singlets [LZ24 (8); LZ25 eq. after (3)], normalised  $\text{tr} = \frac{1}{N} \text{Tr}$ ,  $X \rightarrow X/\sqrt{N}$ , so that  $\langle \text{tr } O \rangle = O(1)$  [LZ24 (7); LZ25 (3)].

### Level (hierarchy).

$\ell(X) = 1$ ,  $\ell(P) = 2$ ,  $\ell(\psi) = \frac{3}{2}$  (for canonical  $\{\psi, \psi\} = \delta$  fermions), additive over letters; chosen so that  $\ell([H, O]) = \ell(O) + 1$  and  $\ell(\{Q, O\}) = \ell(O) + \frac{1}{2}$  [LZ24 §2.5, (25); LZ25 “Hierarchy”; LM §3.2.1]. A “level- $L$  bootstrap” uses positivity matrices built from operators of level  $\leq L/2$  (so entries have level  $\leq L$ ) and all equality constraints among variables of level  $\leq L$ . “Free variables” = variables left after solving the equalities (LZ24 Table 1; LZ25 Table I).

## 2 The constraint catalogue

| Constraint                                | Statement                                                                                                                                                                                                                                                                                                                        | Source                                                                   |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <b>Inner-product positivity</b>           | $M_{ij} = \langle O_i^\dagger O_j \rangle \succeq 0$ for any finite set $\{O_i\}$ : the Gram matrix of $\{O_i E\rangle\}$ . Uses only Hilbert-space positivity, so fermions and sign problems are irrelevant. Adding operators can only tighten bounds (principal minors).                                                       | HHK (7), (15); LM (1.1)–(1.4); LZ25 (5); TASI (95)–(96)                  |
| <b>Stationarity (EOM)</b>                 | $\langle [H, O] \rangle = 0$ for all $O$ (true in any eigenstate or stationary $\rho$ ). For matrix QM take $O$ single-trace; $[H, \text{tr } O]$ is again single-trace.                                                                                                                                                         | HHK (2), (9); LM (1.5); LZ25 (4); TASI (100)                             |
| <b>Eigenstate condition</b>               | $\langle OH \rangle = \langle HO \rangle = E \langle O \rangle$ : stronger, resolves individual eigenvalues (“archipelago”), but bilinear in $(E, \phi)$ hence non-convex; usable by scanning $E$ . Becomes trivial under large- $N$ factorisation.                                                                              | HHK (5) & fn. [20]; LM (2.14) & fn. 2; LZ24 (57)–(58), Fig. 4; TASI §6.2 |
| <b>Reality / time reversal</b>            | $\langle O_1 \cdots O_n \rangle^* = \langle O_n^\dagger \cdots O_1^\dagger \rangle$ ; with real wavefunctions $\langle \text{tr } O_1 \cdots O_n \rangle = \pm \langle \text{tr } O_n \cdots O_1 \rangle$ (– for an odd number of $P$ ’s). Halves the variables.                                                                 | LM (3.34); LZ24 §2.2(2); LZ25 “Constraints”                              |
| <b>Algebra / canonical form</b>           | (Anti)commutation relations reduce every word to a canonical (e.g. normal-ordered) form; for matrices, cyclicity of $\text{tr}$ produces <i>double traces</i> : $\text{tr } O_1 \cdots O_n = \pm \text{tr } O_2 \cdots O_n O_1$ + double trace.                                                                                  | HHK (14), (S37)–(S38); LZ24 (11)–(12), (29); TASI (101)                  |
| <b>Large-<math>N</math> factorisation</b> | $\langle \text{tr } A \text{tr } B \rangle \rightarrow \langle \text{tr } A \rangle \langle \text{tr } B \rangle$ (the <i>only</i> place $N = \infty$ enters in the Hamiltonian approach). Makes the constraints quadratic in the variables. Justified by planar dominance; can fail for operators sensitive to flat directions. | LZ24 (46) & §4; TASI §6.1–6.2                                            |
| <b>Gauge constraint</b>                   | Singlet state: $\langle \text{tr}(OG) \rangle = 0$ for all $O, G$ the $SU(N)$ generator (e.g. $G = i[X, P] - \Psi\Psi^\dagger - \Psi^\dagger\Psi + 2N$ for one bosonic + one fermionic matrix). Removes an enormous number of variables; without it the Marinari–Parisi SDP was intractable.                                     | HHK (11)–(12); LZ24 (13); LZ25 after (1); LM (3.12)–(3.13), §4           |

| Constraint<br>(cont.)                              | Statement                                                                                                                                                                                                                                                                                                                                                                                                                 | Source                                                     |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| <b>Global symmetry (Ward identities, blocking)</b> | If the state is $G$ -invariant, $\langle O_R \rangle = 0$ unless $R$ is the singlet, and $\langle \bar{O}_{\bar{R}r} O_{Rr} \rangle = \delta_{\bar{R}R} \delta_{rr} a_{\bar{R},R}$ . Positivity of the big matrix reduces to $a_{\bar{R},R} \succeq 0$ block by block; channels are related by crossing kernels ( $6j$ symbols). Conserved charges give $\langle [F, O] \rangle = 0$ and block-diagonalise $M$ by charge. | LZ24 (16)–(24), (32)–(35); LZ25 App. F; LM (3.18) & §3.2.1 |
| <b>Ground-state positivity</b>                     | $N_{ij} = \langle O_i^\dagger [H, O_j] \rangle \succeq 0$ : any perturbation $\sum c_i O_i  \Omega\rangle$ has energy $\geq E_0$ . Zero-temperature limit of the energy–entropy balance inequality $\langle O^\dagger O \rangle \log \frac{\langle O^\dagger O \rangle}{\langle O O^\dagger \rangle} \leq \beta \langle O^\dagger [H, O] \rangle$ . Supplies <i>upper</i> bounds and shrinks peninsulas to islands.       | CGSY (1.5)–(1.6); LZ25 (6); TASI (102)–(104); LM (3.20)    |
| <b>KMS / thermal state</b>                         | For $\rho \propto e^{-\beta H}$ : $\beta C - A^{1/2} \log(A^{1/2} B^{-1} A^{1/2}) A^{1/2} \succeq 0$ with $A = \langle O_i^\dagger O_j \rangle$ , $B = \langle O_j O_i^\dagger \rangle$ , $C = \langle O_i^\dagger [H, O_j] \rangle$ (equivalent to KMS given stationarity); made SDP-representable by a rational relaxation of the matrix logarithm. Trivialises for gauged planar models with traced operators only.    | CGSY (1.6), (2.4)–(2.11), §6                               |
| <b>Supercharge (SUSY vacuum)</b>                   | If $Q \Omega\rangle = 0$ : $\langle QO \rangle = \langle OQ \rangle = 0$ , at large $N$ $\langle \{Q_\alpha, O_\alpha\} \rangle = 0$ for fermionic $O$ . Implies ground-state positivity: $\langle O^\dagger [H, O] \rangle = \frac{1}{8} \langle [Q, O]^\dagger [Q, O] \rangle \geq 0$ . The weaker $\langle [Q^\dagger Q, O] \rangle = 0$ was found redundant with EOM.                                                 | LZ24 (14), App. A (47)–(54); TASI (128); LM (3.19)         |
| <b>Nonlinear relaxation</b>                        | Quadratic terms $x_i x_j$ (from factorisation) $\rightarrow$ new variables $\mathcal{Q}_{ij}$ with $\begin{pmatrix} 1 & \vec{x}^T \\ \vec{x} & \mathcal{Q} \end{pmatrix} \succeq 0$ and $M \succeq \mathcal{Q}$ (covariance positivity). Rigorous, convex, weaker than the equality.                                                                                                                                      | LZ25 App. B (B6)–(B7); TASI (98)–(99)                      |

**Objective.** Minimise (maximise)  $\phi(H)$  for a rigorous lower (upper) bound on  $E_0$  (top of spectrum); or fix  $E$  and extremise a one-point function to map the allowed region (“peninsula” if unbounded in some direction, “island” if compact). Typical plots:  $E$  vs  $\langle \text{tr } X^2 \rangle$  (LZ25 Figs. 1–2; LZ24 Figs. 1–2).

### 3 Mechanics of a run

1. Enumerate variables (single-trace singlet words up to level  $L$ ; or, at finite  $N$ , all canonical words).
2. Generate and *symbolically solve* the linear equalities (algebra, reality, gauge, Ward, EOM, supercharge) for a set of free variables. This step, not the SDP, dominates cost (LZ24 §3; LM §3.2.2: 8 h of algebra vs 5–10 min per SDPB point).
3. Build the positivity matrices ( $M$ ,  $N$ , relaxation blocks) as linear matrix inequalities in the free variables, one block per (charge, irrep).
4. Solve the SDP. Generic solvers (Mathematica, SCS) suffice at low level; the problems lack strict feasibility, so high level needs arbitrary precision: SDPA-GMP (LZ25 App. A) or SDPB (LM). Bounds should tighten monotonically with level; plateaus followed by jumps happen (LM Fig. 13 and §3.2.2).
5. Why it converges: if  $\phi(\mathbf{1}) = 1$ ,  $\phi(O^\dagger) = \phi(O)^*$  and  $\phi(O^\dagger O) \geq 0$  for *all*  $O$ , then  $\phi = \text{Tr } \rho(\cdot)$  for a genuine state; adding  $\phi(OH) = E\phi(O)$  makes it an eigenstate (HHK fn. [20]). For a finite fermionic Hilbert space “all  $O$ ” is a finite set, so the bootstrap is exact at the top level.
6. Sanity checks: (a) feed the exact state (ED or known solution) into every constraint and verify  $M \succeq 0$ , EOM residual 0; (b) reproduce analytic low-level bounds, e.g. the virial/Polchinski bounds (TASI (114)–(127); LZ25 (7), App. E).

### 4 Worked micro-examples

(a) **Fermion number and  $E \geq 0$  from a  $2 \times 2$  block** (LM (3.26)–(3.28)): with  $\{\Psi, \Psi^\dagger\}$  operators,  $\begin{pmatrix} \langle \text{tr } \Psi^\dagger \Psi \rangle & \langle \text{tr } \Psi^\dagger \Psi^\dagger \rangle \\ \langle \text{tr } \Psi \Psi \rangle & \langle \text{tr } \Psi \Psi^\dagger \rangle \end{pmatrix} \succeq 0$

gives  $0 \leq F \leq N^2$ ; combined with the bosonic block it yields  $\langle H \rangle \geq 0$  with  $E = 0$  only at  $F = 0$  for the quadratic superpotential.

(b) **Level-5 bosonic Yang–Mills** (LZ25 App. E): 19 variables, equalities (E4)–(E32) leave 3 free; irrep blocks (E35)–(E38) give the analytic peninsula/island (E39).

(c) **SUSY sum of squares** ( $\star$ , this project): for  $H = \{Q, \bar{Q}\}$  include  $Q, \bar{Q}$  among the  $O_i$ . Then  $\phi(\bar{Q}Q)$  and  $\phi(Q\bar{Q})$  are diagonal entries of  $M \succeq 0$ , so  $\phi(H) \geq 0$  is *automatic*; the true content of a lower bound in a lifted sector is whatever forces  $\phi(\bar{Q}Q) + \phi(Q\bar{Q}) > 0$ .

(d) **Representation-resolved bounds by hand** (KMPT §3.1–3.2):  $C_2 \pm \frac{1}{2}A \cdot A = \frac{1}{4}(A \pm A)^2 \geq 0$  and Cauchy–Schwarz with a symmetry-averaged density matrix  $\rho_R = \frac{1}{\dim R} \sum_i |e_i\rangle\langle e_i|$  give  $E$ -vs-Casimir bounds (3.8), (3.21) that are saturated in Casimir-type fermionic matrix models. This is a two-operator bootstrap in each irrep.

## 5 Specialisation to purely fermionic matrix models

Setting:  $p$  complex fermion matrices  $\Psi_{ij}^{(f)}$ ,  $\{\Psi_{ij}, \bar{\Psi}_{kl}\} = \delta_{il}\delta_{jk}$ ,  $\bar{\Psi}_{ij} = \Psi_{ji}^\dagger$ ,  $Q$  cubic in  $\Psi$ ,  $H = \{Q, \bar{Q}\}$ , R-charge  $N_\Psi = \text{Tr } \Psi \bar{\Psi}$ ,  $[N_\Psi, Q] = 3Q$  [Chen (1.1)–(1.2), (2.4)]. Hilbert space  $2^{pN^2}$ -dimensional.

- **Pauli truncation.** Each mode appears at most once, so the algebra of words closes at finite length; at fixed  $N$  the constraint set is finite and the bootstrap can be *exact* at the top level. No flat directions, no continuum of moments.
- **Blocks.**  $M$  is block-diagonal in  $N_\Psi$  (as in LM §3.2.1:  $F = 0, \pm 1, \pm 2$  blocks) and in  $SU(N) \times \text{flavor}$  irrep. Fix a sector with  $\phi(N_\Psi) = k$ ,  $\phi(N_\Psi^2) = k^2$ : since  $\phi((N_\Psi - k)^2) = 0$ , positivity forces  $\phi((N_\Psi - k)O) = 0$  for every  $O$  in the set. Extremising  $\phi(H)$  then bounds the sector ground energy  $E_0(k)$  ( $\star$ ).
- **Which sector is the ground state.** For  $Q = \text{Tr } \Psi^3$ ,  $H = 3N(N^2 - 1) - 9\hat{C}_2$  [Chen (2.8)]: singlets are the *top* of the spectrum and the BPS states form the maximal-Casimir irrep  $r_*$  [Chen (2.18)–(2.22)]. A gauged (singlet) bootstrap bounds the wrong end. Work ungauged, with irrep-resolved blocks; the singlet-state justification for using open-index operators (LZ25 after (6)) must be replaced by the irrep-averaged functional  $\rho_R$  of KMPT (3.15).
- **SUSY floor and how to lift it** ( $\star$ ). With  $Q, \bar{Q}$  in the basis,  $\phi(H) \geq 0$  is free. In a lifted sector the SDP must additionally learn that annihilators act first on the sector ground state, e.g.  $\phi(Q\bar{Q}) = 0$  at  $N_\Psi = 0$ . In a basis of elementary letters this follows from  $\phi(N_\Psi) = 0$ ,  $\phi(\Psi_{ij}\bar{\Psi}_{ji}) \geq 0$  and Cauchy–Schwarz. In a basis of *gauge-invariant single traces only* it does not: at  $N = 3$ , levels 3 and 4, all lifted sectors of the single-matrix model return the trivial bound 0 (verified 2026-09-15). Open-index (adjoint-valued) operators are required in the positivity matrix; the precedent is the ungauged-MQM bootstrap of CGSY §3.1, whose adjoint Gram matrix is decomposed into  $U(N)$  tensor structures (3.5) with no gauge constraint imposed.
- **Ground-state positivity in a sector.**  $\langle O^\dagger[H, O] \rangle \geq 0$  is valid for the sector ground state only if  $O$  preserves the sector ( $[O, N_\Psi] = 0$ , and likewise for the irrep); a charge-changing  $O$  may lower the energy by leaving the sector. For a genuine SUSY vacuum ( $E = 0$ ) it is implied by the supercharge constraints (LZ24 App. A) and adds nothing.
- **Concentration as a bootstrap target** [CCSY Conj. 1, §2; Chen (2.22)]. R-charge concentration = one BPS degree per irreducible complex (labels:  $N_\Psi \bmod 3$ ,  $SU(N)$  irrep, flavor). A rigorous  $E_0(k, \mathbf{r}) > 0$  outside a window of  $\leq 3$  consecutive  $k$  proves it; the single-matrix baseline has  $E_0 = 0$  on  $N + 1$  consecutive sectors.
- **Exact single-trace  $H$  for  $Q = \text{Tr } \Psi^3$**  ( $\star$ , docs/derivations.md D1):  $H = \frac{9}{2}(\text{Tr}[\Psi\Psi\bar{\Psi}\bar{\Psi}] + \text{Tr}[\bar{\Psi}\bar{\Psi}\Psi\Psi]) - \frac{3}{2}N(N^2 - 1)$ , equivalently  $\hat{C}_2 = NN_\Psi - \text{Tr } \Psi \text{Tr } \bar{\Psi} - \text{Tr}[\Psi\Psi\bar{\Psi}\bar{\Psi}]$ . (An earlier fitted constant  $-9(N - 1)^2$  is wrong for  $N \geq 4$ .)

## 6 Numbers to calibrate against

- Bosonic YM matrix QM (LZ25 Table II):  $D=2$ ,  $M=0$ :  $E \in [0.707832, 0.707868]$  at level 14;  $D=9$ :  $E \in [6.69946, 6.69968]$ ,  $\langle \text{tr } X_I X_I \rangle \in [2.29195, 2.29218]$  at level 11 (Monte Carlo 6.695(5), 2.291(1)). Islands first appear at level 10 for  $M = 0$ .
- BFSS (LZ24 Table 2):  $\langle \text{tr } X^2 \rangle \geq 0.355$  at level 9 (Monte Carlo  $\approx 0.378$ ); analytic level 6:  $\frac{3}{4}(3/50)^{1/3} \approx 0.2936$ .
- Marinari–Parisi (LM §3.2.2): level 8,  $44 \times 44$ , 27 free variables;  $E_0/N^2 \approx 0.196 g^{2/3}$  (WKB 0.242); no upper bound obtained.
- Single-matrix fermionic model (Chen; this project):  $E \in \{0, 18\}$  ( $N=2$ ),  $\{0, 18, 45, 72\}$  ( $N=3$ );  $C_2^{\max} = N(N^2 - 1)/3$ ;  $\dim r_* = 3^{N(N-1)/2}$ ; BPS window  $[N(N-1)/2, N(N+1)/2]$ . Three-matrix model at  $N = 2$ : 972 BPS states at  $N_\Psi \in \{5, 6, 7\}$ , counts 243:486:243, 168 distinct energies (ED, reproduced 2026-09-15). Compare the  $\hat{q} = 3$  super-Schwarzian/SYK prediction of three charges in ratio  $\cos(\pi k/3) = 1:2:1$  (TW [10] (3.10); FGMS [11] (5.7)) and the TW chaos test:  $r$ -ratio of singular values of  $Q_k$  per charge pair,  $\beta = 2$ .

## References

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